Differential Abundance Testing w Hooke

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SASC meeting

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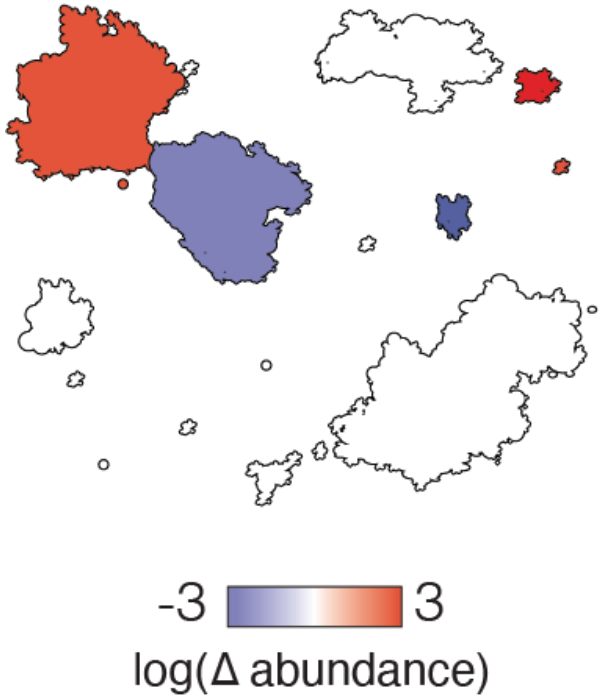
Motivation: single-cell experiments across multiple conditions

- Single-cell data lets us measure cell states at high resolution
- Many experiments involve perturbations:
  - CRISPR screens
  - drug treatments
  - disease vs control



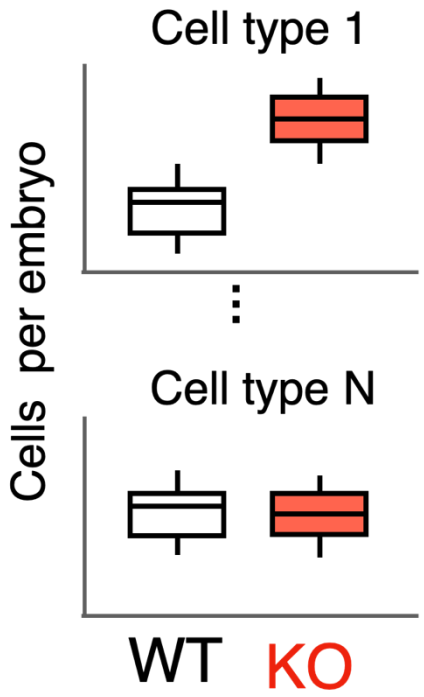
Which cell populations change in response to a perturbation?

What is differential abundance?

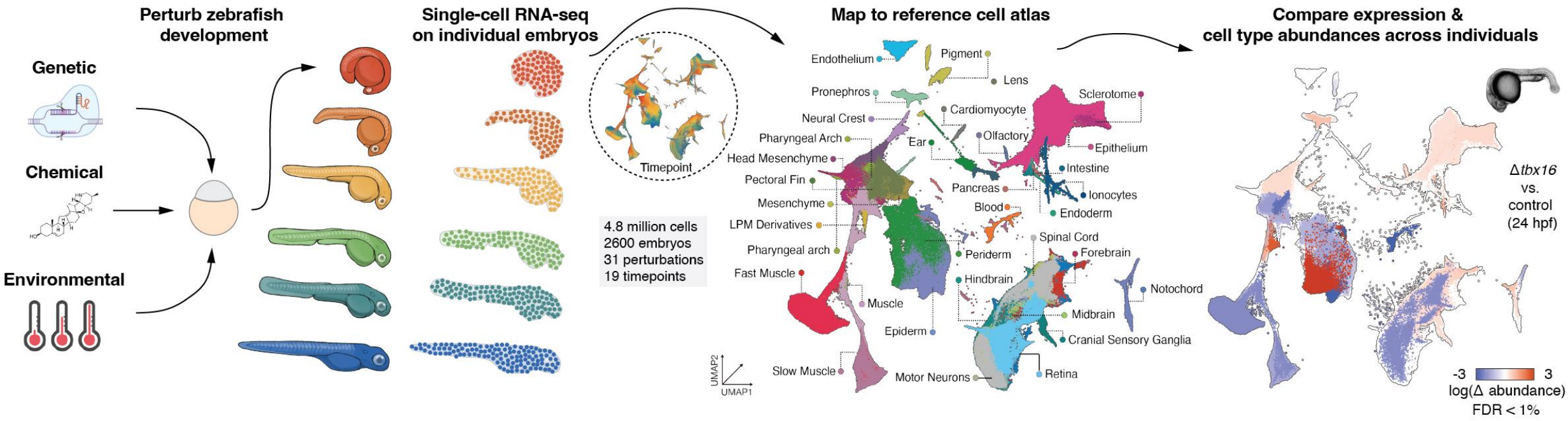
Differential expression:
changes in genes

Differential abundance:
changes in cell populations

wildtype	mutant
gene 1	*gene 1* ↓
gene 2	*gene 2*
gene 3	*gene 3*
gene 4	*gene 4*
gene 5	*gene 5* ↓
gene 6	*gene 6* ↓
gene 7	*gene 7* ↑



The Trapnell Lab uses Hooke to analyze perturbations in zebrafish development



Data input requirements:

Hooke is built for experiments with multiple treatments and replicates

	cells						
gene 1	0	1	4	0	2	1	1
gene 2	2	0	2	2	1	0	1
gene 3	1	2	0	0	1	0	1
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.
gene i	x	x	x	x	x	x	x

gene count matrix

	sample			
cell type 1	5	4	1	1
cell type 2	24	17	20	18
cell type 3	8	11	4	9
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
cell type i	x	x	x	x

cell count matrix

Data input requirements:

Metadata (covariates) for each individual (e.g. condition, time, batch)

cell count matrix

	individual			
cell type 1	5	4	1	1
cell type 2	24	17	20	18
cell type 3	8	11	4	9
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
cell type i	x	x	x	x

covariates

		time	perturbation	expt
individual	1	18	WT	1
	2	18	tbx16	1
	3	24	WT	2
	4	24	tbx16	2

Challenges in modeling differential abundance

- Cell populations are **not independent**
- Counts vary due to:
  - biology
  - sampling noise
  - experimental factors

Want:

Joint model of cell counts

Accounts for covariates and variability

Chiquet, J. *et al*. ICML (2021).
Aitchison, J. & Ho, C. Biometrika (1989).

Approach: Poisson-Lognormal models

- multivariate mixed generalized linear model with Poisson distribution
- variational approximation circumvents computational issues
- can easily model covariates + sampling

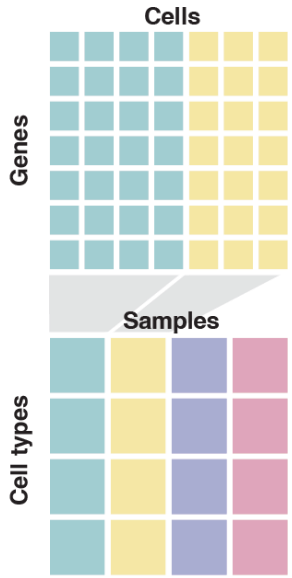
Chiquet, J. *et al*. ICML (2021).
Aitchison, J. & Ho, C. Biometrika (1989).

Hooke quantifies the effects of perturbations of cells in different molecular states

Poisson log-normal (PLN) models

Input:
Single-cell count matrices

Aggregate cells by sample and cell type:



Implementation:

latent layer

latent variable *mean* *covariance*

$$\mathbf{Z_i} \sim \mathcal{N}(\mathbf{x_i}^\top \mathbf{\Theta}, \mathbf{\Sigma})$$

covariates *d x n* — *regression parameters* *d x p*

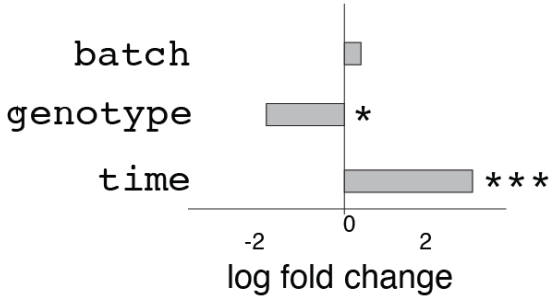
observation layer

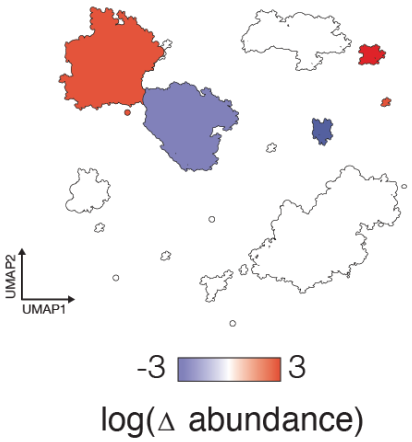
$$\mathbf{Y_i} | \mathbf{Z_i} \sim \mathcal{P}(\exp\{\mathbf{o_i} + \mathbf{Z_i}\})$$

cell type counts *n x p* — *Poisson* — *offset* *n x p*

Output:

Effects on proportions of cell states from time, genetic perturbation, etc.





What this tutorial will cover:

- How to prepare input data
- How to set up a Hooke model
- How to plot and interpret results